1. Arp (linux.cs.pdx.edu)

IPV4 Address: 131.252.208.103/24
Hardware Address: 52:54:00:13:a0:c6

What is the default router's IP address (e.g. the gateway address for the default route 0.0.0.0/0)

131.252.208.1

What is the name of the default router and its hardware address?

router.seas.pdx.edu
00:00:5e:00:01:01

How many entries are there in the ARP table?

15

List any IP addresses that share the same hardware address

```
Duplicate hardware address: at IP: (131.252.208.53)
Duplicate hardware address: at IP: (131.252.208.83)
Duplicate hardware address: at IP: (131.252.208.84)
Duplicate hardware address: at IP: (131.252.208.138)
Duplicate hardware address: at IP: (131.252.208.21)
Duplicate hardware address: at IP: (131.252.208.118)
Duplicate hardware address: at IP: (131.252.208.48)
Duplicate hardware address: at IP: (131.252.208.55)
Duplicate hardware address: at IP: (131.252.208.23)
Duplicate hardware address: at IP: (131.252.208.20)
Duplicate hardware address: at IP: (131.252.208.121)
Duplicate hardware address: at IP: (131.252.208.34)
Duplicate hardware address: at IP: (131.252.208.137)
Duplicate hardware address: at IP: (131.252.208.28)
Duplicate hardware address: at IP: (131.252.208.85)
Duplicate hardware address: at IP: (131.252.208.54)
Duplicate hardware address: at IP: (131.252.208.81)
Duplicate hardware address: at IP: (131.252.208.172)
Duplicate hardware address: at IP: (131.252.208.77)
Duplicate hardware address: at IP: (131.252.208.17)
Duplicate hardware address: at IP: (131.252.208.110)
Duplicate hardware address: at IP: (131.252.208.43)
Duplicate hardware address: at IP: (131.252.208.63)
```

How many less hardware addresses are there than IP addresses in the ARP table?

There are one less hardware addresses than IP addresses

Include the command in your lab notebook

arp -an | awk -F '([()])' '{print $2}' > arp_entries

What network prefix do most of the IP addresses in the ARP table share?

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2. Arp (Cloud)

Include both in your lab notebook ip and hardware address

```
2: ens4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1460 qdisc mq state UP group default qlen 1000
    link/ether 42:01:0a:8a:00:02 brd ff:ff:ff:ff:ff:ff
    inet 10.138.0.2/32 metric 100 scope global dynamic ens4
       valid_lft 86233sec preferred_lft 86233sec
    inet6 fe80::4001:aff:fe8a:2/64 scope link
       valid_lft forever preferred_lft forever
```

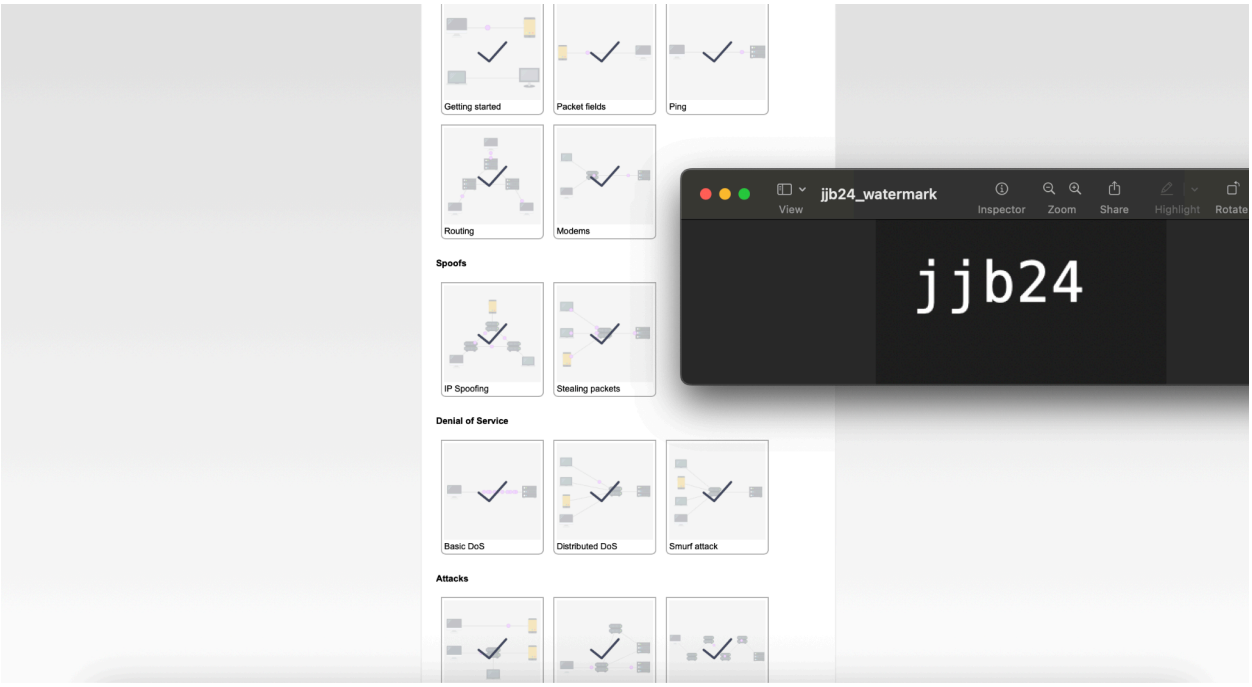
ip: 10.138.0.2/32
hardware: 42:01:0a:8a:00:02

What is the default router's IP address (e.g. the gateway address for the default route 0.0.0.0/0)
10.138.0.1

What is the default router's hardware address?
42:01:0a:8a:00:01

NetSim

All levels complete



1.3 cloud networking

Show a screenshot of the output for the scan for your lab notebook.

```
jjb24@course-vm:~$ nmap 10.128.0.2 10.128.0.3 10.128.0.4
Starting Nmap 7.80 ( https://nmap.org ) at 2025-10-05 19:55 UTC
Nmap scan report for instance-20251005-193933.c.cloud-brown-jjb24.internal (10.128.0.2)
Host is up (0.032s latency).
Not shown: 999 closed ports
PORT   STATE SERVICE
22/tcp open  ssh

Nmap scan report for instance-20251005-194033.c.cloud-brown-jjb24.internal (10.128.0.3)
Host is up (0.032s latency).
Not shown: 999 closed ports
PORT   STATE SERVICE
22/tcp open  ssh

Nmap scan report for instance-20251005-194133.c.cloud-brown-jjb24.internal (10.128.0.4)
Host is up (0.032s latency).
Not shown: 999 closed ports
PORT   STATE SERVICE
22/tcp open  ssh

Nmap done: 3 IP addresses (3 hosts up) scanned in 1.26 seconds
```

How many subnetworks are created initially on the `default` network? How many regions does this correspond to? (Use a pipe to pass output to `grep` in order to return specific lines of output and then another to pass output to `wc` to count them: `| grep default | wc -l`)

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Given the CIDR prefix associated with each subnetwork, how many hosts does each subnetwork support?

20

Which CIDR subnetworks are these instances brought up in? Do they correspond to the appropriate region based on the prior commands?

10.128.0.2

10.128.0.2

Yes, they do correspond to the appropriate regions.

Perform a ping to instance 2, take a screenshot of this:

```
PING 10.128.0.5 (10.128.0.5) 56(84) bytes of data.
64 bytes from 10.128.0.5: icmp_seq=1 ttl=64 time=32.6 ms
64 bytes from 10.128.0.5: icmp_seq=2 ttl=64 time=31.9 ms
64 bytes from 10.128.0.5: icmp_seq=3 ttl=64 time=31.9 ms
64 bytes from 10.128.0.5: icmp_seq=4 ttl=64 time=32.0 ms
64 bytes from 10.128.0.5: icmp_seq=5 ttl=64 time=31.9 ms
64 bytes from 10.128.0.5: icmp_seq=6 ttl=64 time=31.9 ms
64 bytes from 10.128.0.5: icmp_seq=7 ttl=64 time=32.0 ms
64 bytes from 10.128.0.5: icmp_seq=8 ttl=64 time=32.0 ms
64 bytes from 10.128.0.5: icmp_seq=9 ttl=64 time=32.1 ms
64 bytes from 10.128.0.5: icmp_seq=10 ttl=64 time=32.0 ms
```

- **From the figure in the previous step. What facilitates this connectivity: the virtual switch or the VPN Gateway?**
  the VPN gateway facilitates this connectivity

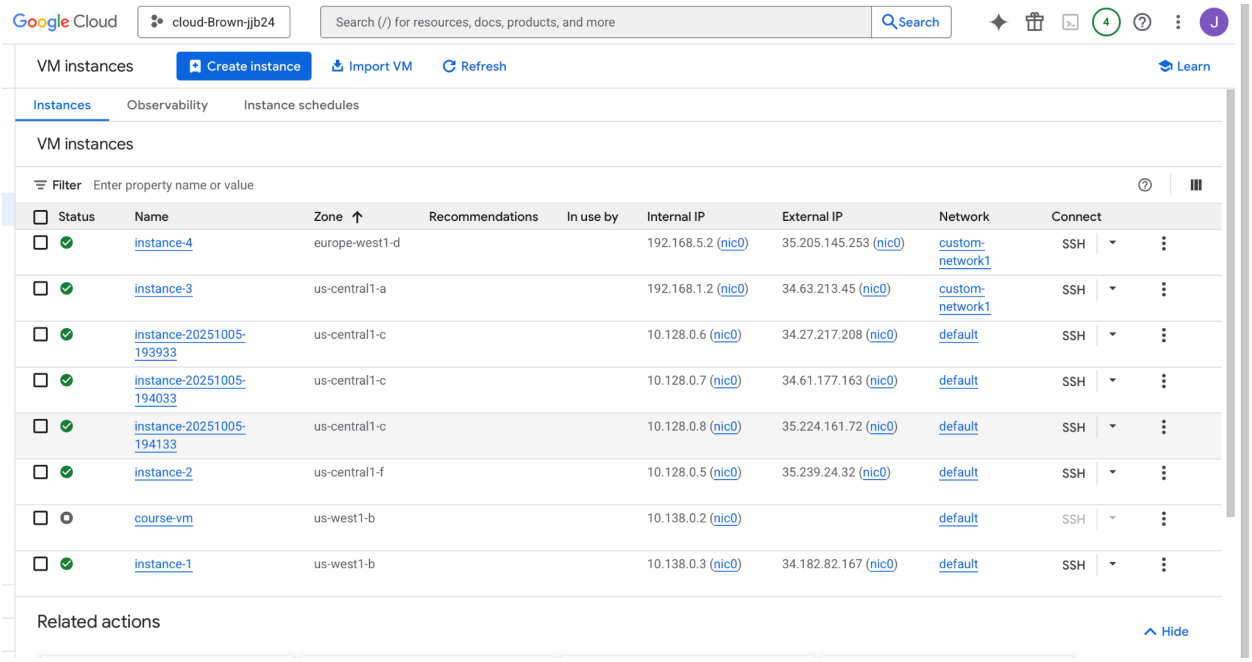
Take a screenshot of the new subnets created in `custom-network1` alongside the default subnetworks in those regions assigned to the `default` network.

```
EXTERNAL_IPV6_PREFIX:
jjb24@cloudshell:~ (cloud-brown-jjb24)$ gcloud compute networks subnets list --regions=us-central1,europe-west1
NAME: default
REGION: europe-west1
NETWORK: default
RANGE: 10.132.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:

NAME: subnet-europe-west-192
REGION: europe-west1
NETWORK: custom-network1
RANGE: 192.168.5.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:

NAME: default
REGION: us-central1
NETWORK: default
RANGE: 10.128.0.0/20
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
INTERNAL_IPV6_PREFIX:
EXTERNAL_IPV6_PREFIX:

NAME: subnet-us-central-192
REGION: us-central1
NETWORK: custom-network1
RANGE: 192.168.1.0/24
STACK_TYPE: IPV4_ONLY
IPV6_ACCESS_TYPE:
```

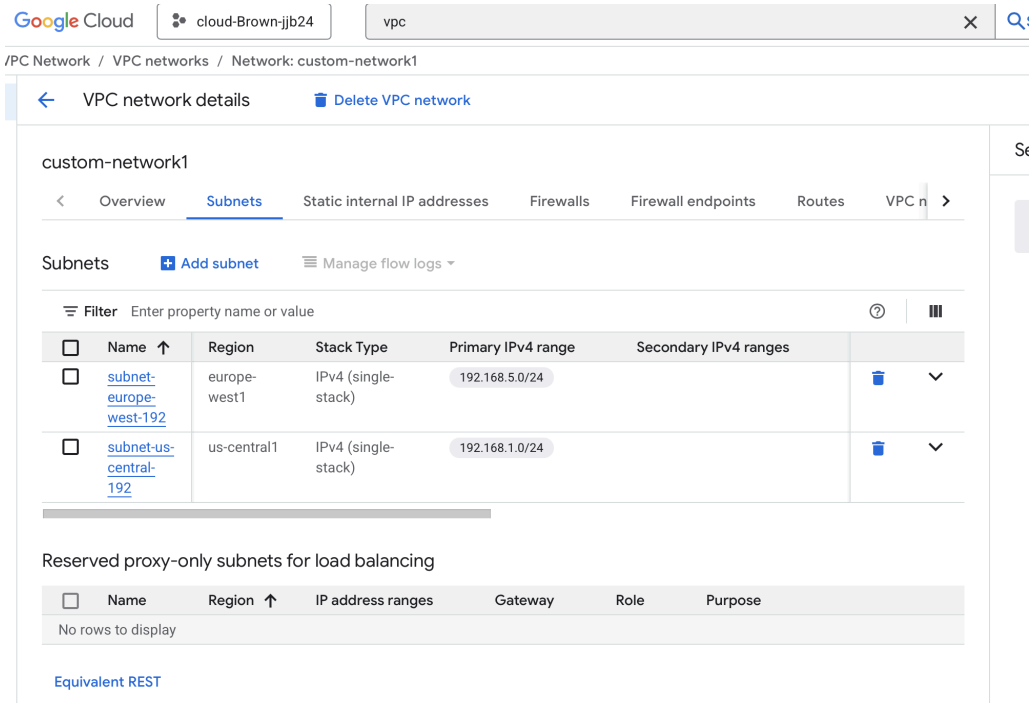
Explain why the result of this ping is different from when you performed the ping to `instance-2`.

The reason why the result of the ping from this time is different from the result of the original ping is due to the vpn gateway

Take screenshots of all 4 instances in the UI including the network they belong to.
(next page)



Take a screenshot of the subnetworks created for the custom-network1 network and some of the subnetworks of the default network showing their regions, internal IP ranges and Gateways.



Google Cloud | cloud-Brown-jjb24 | vpc | Search

VPC Network / VPC networks / Network: default

VPC network details | Delete VPC network

default

Overview | Subnets | Static internal IP addresses | Firewalls | Firewall endpoints | Routes | VPC

Subnets | Add subnet | Manage flow logs

Filter | Enter property name or value

Name	Region	Stack Type	Primary IPv4 range	Secondary IPv4 ranges
default	africa-south1	IPv4 (single-stack)	10.218.0.0/20	
default	asia-east1	IPv4 (single-stack)	10.140.0.0/20	
default	asia-east2	IPv4 (single-stack)	10.170.0.0/20	
default	asia-northeast1	IPv4 (single-stack)	10.146.0.0/20	
default	asia-northeast2	IPv4 (single-stack)	10.174.0.0/20	
default	asia-northeast3	IPv4 (single-stack)	10.178.0.0/20	
default	asia-south1	IPv4 (single-stack)	10.160.0.0/20	
default	asia-south2	IPv4 (single-stack)	10.190.0.0/20	

Select a subn

Please